STATISTICS · CHAPTER D2–D3

Descriptive Statistics

A plain-English study guide with worked examples

This guide rebuilds every idea in your *Descriptive Statistics* slides from the ground up: what each concept **means**, the **formula** with its intuition, a **worked example** using the exact numbers from your course, and a **"How to think in the exam"** box for the kind of question that gets asked. Part I describes a **single** quantitative variable; Part II describes the **relationship between two** variables.

Needle graph & histogram | Mean · variance · SD | Median · quartiles · IQR | EDF & percentiles | Boxplot & outliers | Log & z-score transforms | Binned data | Contingency tables | Side-by-side boxplots | Scatterplot · covariance · correlation | Linear combinations

How to use this guide — the big picture

Descriptive statistics has exactly one job: **take a pile of numbers and describe it** so a human can understand it — its typical value, how spread out it is, its shape, and (in Part II) how two variables move together. Every tool below is just a different lens on those questions.

THE ONE DECISION THAT DRIVES EVERYTHING

Before you pick any graph or measure, ask: **what type is my variable?**

- **Qualitative (categorical)** — categories, e.g. *handiness*, *region*. You can only count.
- **Quantitative** — numbers you can average. Split further into **discrete** (countable: household size) and **continuous** (measured: income, area).

The type tells you which graph, which centre, and which spread measure are even *allowed*.

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PART I

Describing one quantitative variable

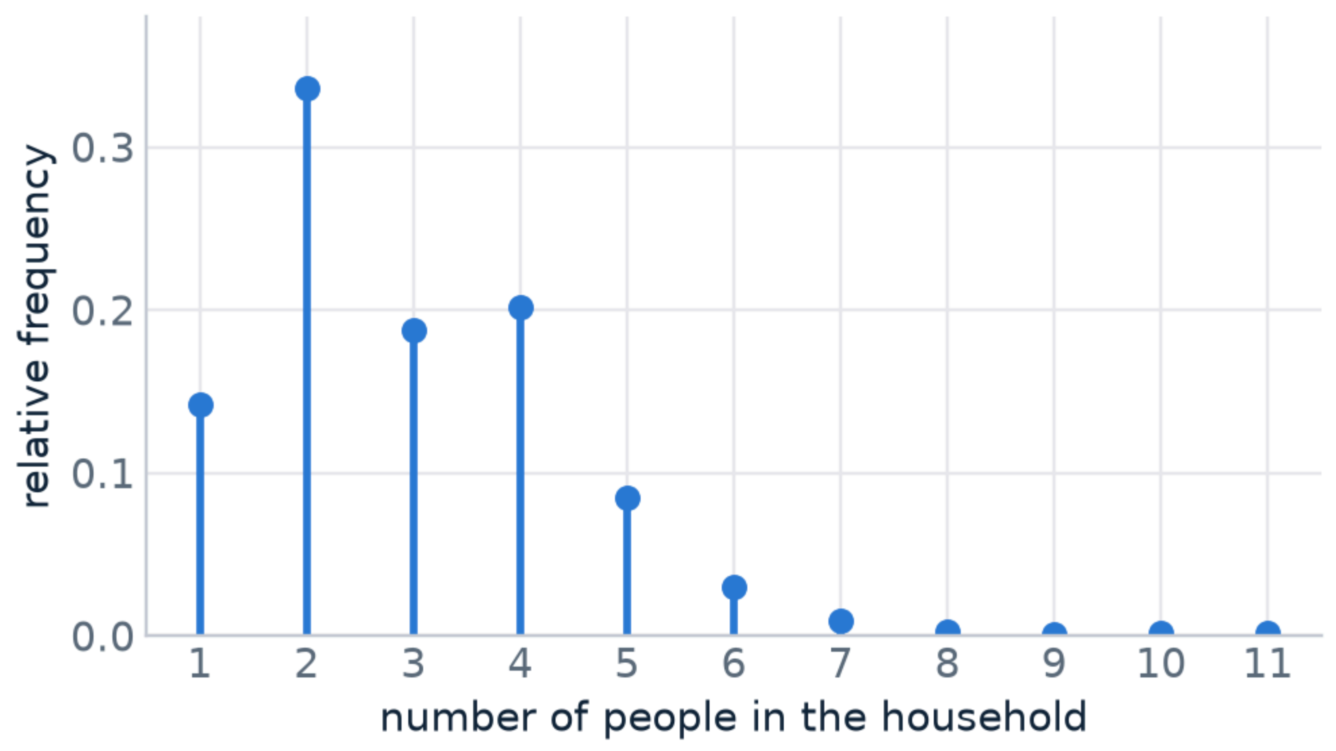
SECTION 1 · SLIDE SET D2.1

The distribution: needle graph & histogram

The **distribution** of a variable is simply the answer to “*which values occur, and how often?*”. Exactly as with ordinal qualitative variables, you can build it from frequencies: **a.f.** (absolute frequency = raw count), **r.f.** (relative frequency = count ÷ n = proportion), and **c.r.f.** (cumulative relative frequency). The right *picture* depends on how many distinct values there are.

1.1 Discrete variable → needle graph

When a variable takes only a handful of distinct values (household size = 1, 2, 3, …), draw a **needle graph**: one vertical “needle” per value, its height = frequency (or relative frequency). Unlike a bar chart for categories, the needles sit on a genuine number line, so keep the **numeric scale correct** and don’t leave category-style gaps.



Needle graph of household size (ESS Belgium). Each needle’s height is the relative frequency of that value.

HOW TO READ A DISTRIBUTION — 4 QUESTIONS

Whatever the graph, describe the **shape** with these four checks. They come up verbatim in exam questions:

1. **Symmetry:** symmetric, **skewed to the right** (long tail on the high side), or **skewed to the left**?
2. **Modality: unimodal** (one peak), **bimodal** (two), or **multimodal**?
3. **Outliers:** any values sitting far from the rest?
4. **Centre & spread:** roughly where is the bulk, and how wide is it?

The four shapes you must be able to name on sight. “Skew” always points in the direction of the *long tail*.

1.2 Continuous (or discrete-with-many-values) → group into bins

For a continuous variable, almost every observation is unique, so a needle graph becomes a useless forest of height-1 needles. Instead we **group outcomes into bins (classes)** and count how many land in each bin — a **frequency table**. The graph of a binned distribution is a **histogram**.

CHOOSING BINS — RULES OF THUMB

The number of bins is a **trade-off**: too few loses information, too many is spiky and unreadable. Guidelines from the slides:

- Aim for roughly **5 to 20** bins; a common rule is ≈ √**n** bins (Sturges’ rule is another).
- Bin width ≈ range ÷ number of bins; first bin starts a little below the minimum.
- Avoid one giant bin holding almost all the data — that hides the shape.

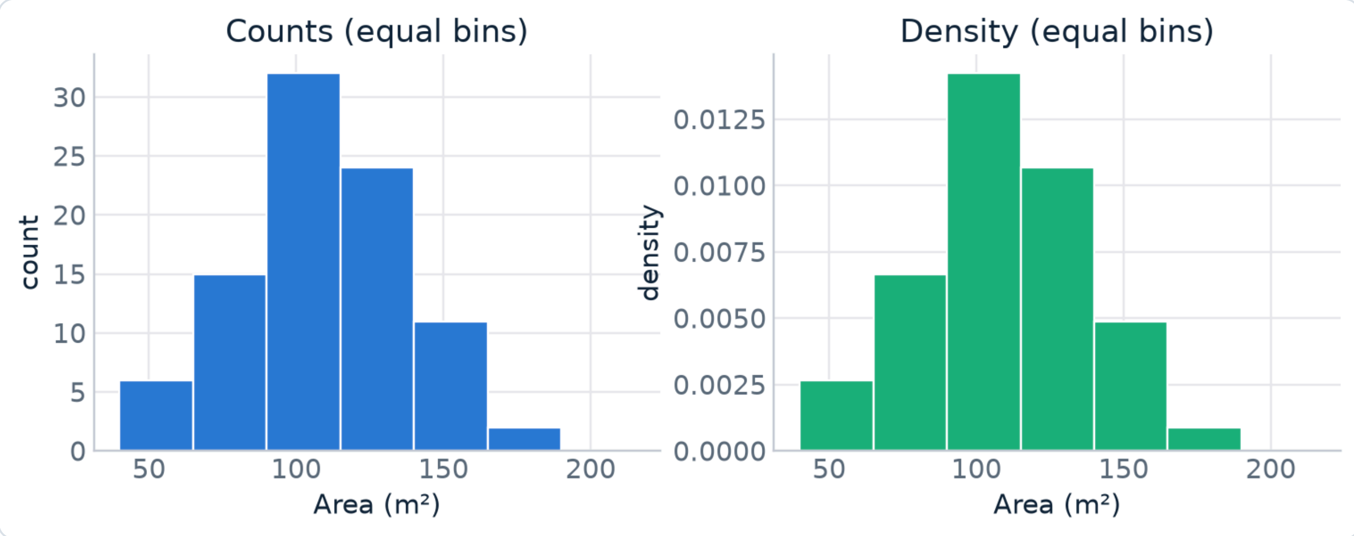
1.3 The y-axis choice — and why *density* matters

A histogram’s **x-axis** is always the binned outcome. The **y-axis** has three options, and picking the wrong one is a classic trap:

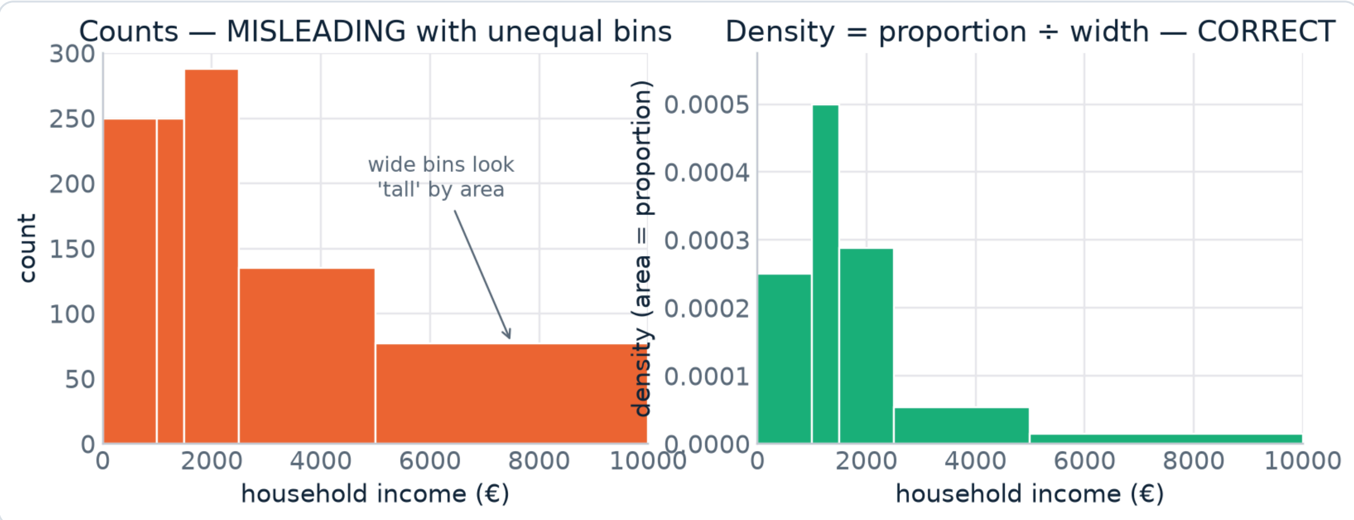
y-axis	bar height means	safe when…
Counts	how many observations in the bin	bins are **equal** width
Proportions	count ÷ n	bins are **equal** width
Density = proportion ÷ width	**area = proportion**	**always (esp. unequal bins)**

In a **density histogram** it is the **area** on top of a bin — not its height — that equals the bin’s proportion, and the **total area under the histogram is exactly 1**. The height is

$$h_i = \frac{\hat{p}_i}{b_i} = \frac{\text{proportion in bin } i}{\text{width of bin } i}$$



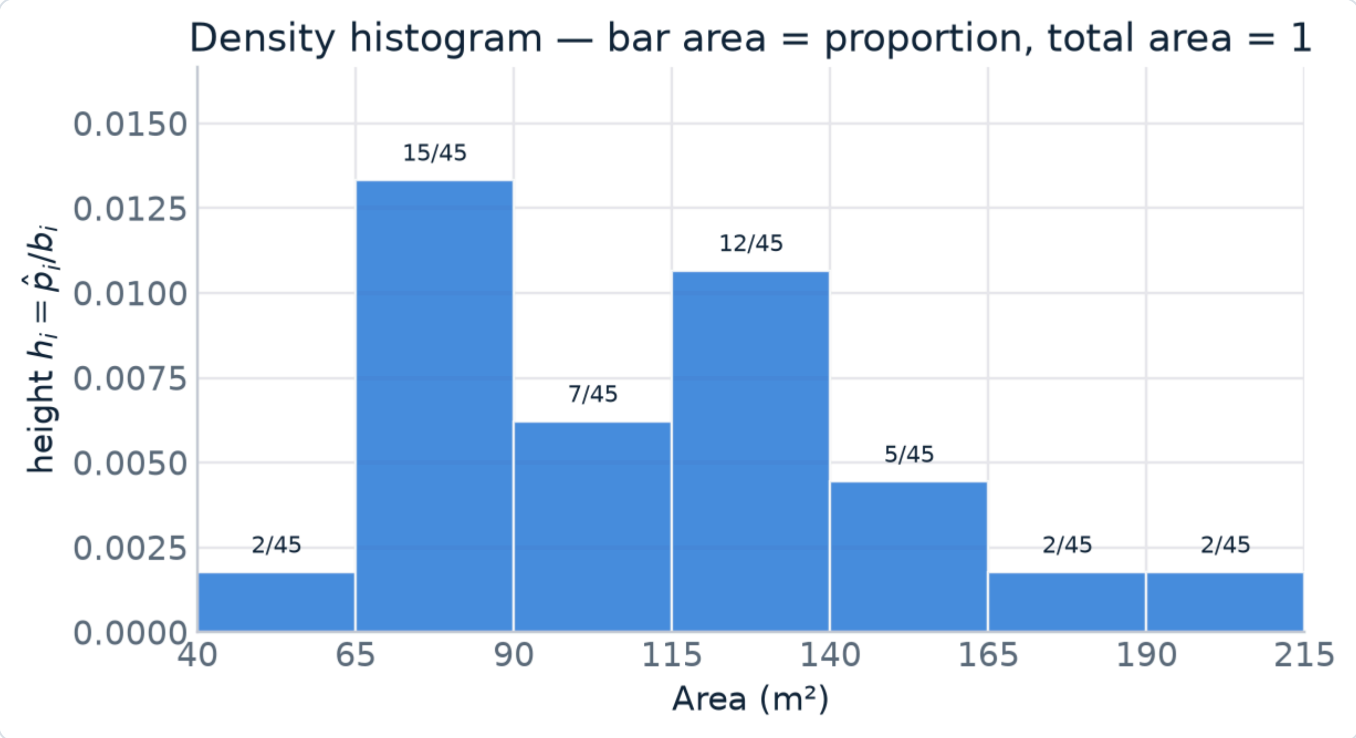
With **equal** bin widths, counts / proportions / density all have the *same shape* — any y-axis is fine.



With **unequal** bins (household income), the **counts** histogram is misleading — wide bins look tall just because they cover more ground. Dividing by width restores honest areas.

WORKED EXAMPLE · APARTMENT AREA (M²)

Bins of width 25 with counts 2, 15, 7, 12, 5, 2, 2 ($n = 45$). The proportion for the [65;90) bin is $15/45 = 0.333$, so its density height is $0.333/25 = \mathbf{0.0133}$. Do that for every bin and the bars' **total area = 1**. That is what the figure below is showing — the `nᵢ/45` labels are the proportions each bar's area represents.



Density histogram: each bar's **area** is that bin's proportion; the heights are proportion ÷ width.

HOW TO THINK IN THE EXAM

- "Draw / interpret the histogram" with **unequal bins given** → they want the **density** (height = proportion ÷ width). Compute a table with columns n_i, $\hat{p}_i$, b_i, h_i.
- "What proportion of observations fall in this range?" on a density histogram → answer with **area** (height × width), never height.
- "Describe the distribution" → run the 4-question checklist and use the words *unimodal / skewed-right / no clear outliers*. The ESS *social-trust* example is "unimodal, slightly skewed to the left, no outliers."
- Don't label a value an **outlier too quickly** — a long tail (income, age) is skew, not necessarily outliers.

SECTION 2 · SLIDE SET D2.2

Summary measures: centre & spread

A graph shows the whole shape; **summary measures** compress it into a few numbers. You always report **two** things together — a measure of **centre** (where is the middle?) and a measure of **spread** (how dispersed?). Reporting one without the other is the classic mistake in the quote below.

> **WHY YOU MUST DISCUSS BOTH**
>
> *"A statistician who could not swim walked confidently into a river that was **on average** 1 metre deep. He drowned."* — the mean hid a deep middle. **Centre + spread, always together.**

2.1 The mean (average)

For a sample $x_1, \dots, x_n$:

$$\bar{x} = \frac{1}{n} (x_1 + \cdots + x_n) = \frac{1}{n} \sum_{i=1}^{n} x_i$$

For a **discrete** variable with a few outcomes $m_1, \dots, m_k$ occurring $n_1, \dots, n_k$ times, the same thing is a **weighted average** — weight each outcome by how often it occurs:

$$\bar{x} = \frac{1}{n} (n_1 m_1 + \cdots + n_k m_k) = \sum \hat{p}_i \, m_i$$

ESS *social-trust* is scored 0 (no trust) – 10 (complete trust) as the average of three trust items; its mean is exactly this weighted average over the outcomes 0…10.

2.2 Spread: from "average deviation" to variance & standard deviation

Spread should measure **how far observations sit from the mean on average**. The natural first idea — average the **absolute** deviations $|x_i - \bar{x}|$ — works, but the absolute value is mathematically awkward (not differentiable). So statistics squares instead, giving the **variance**:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2 \quad \text{and} \quad s = \sqrt{s^2}$$

The **standard deviation** s is the square root of the variance. (Discrete version: replace the sum by $\sum n_i (m_i - \bar{x})^2$.)

> **TWO THINGS GRADERS LOVE TO TEST HERE**
>
> **(a) Why divide by $n-1$, not n?** Those are the **degrees of freedom**. Once the mean is fixed, only $n-1$ of the deviations are free to vary (the last is forced, because deviations sum to 0). Dividing by $n-1$ makes s^2 an **unbiased** estimator of the population variance.
> **(b) Units.** The variance is in **squared** units (euro²), which is meaningless to interpret; the **standard deviation is back in the original units** (euro). Interpret with s, not s^2.

Interpretation of s: it is **0 only if every observation is identical**; otherwise $s > 0$, and it grows as the data become more dispersed. A useful rule of thumb for roughly bell-shaped data: about **95%** of observations lie within $\bar{x} \pm 2s$ and about

99% within $\bar{x} \pm 3s$.

2.3 Comparing spreads: the variation coefficient

Is a standard deviation "big"? That depends on the scale. The **variation coefficient** makes spread **relative** and unit-free, so you can compare variables measured in different units:

$$\mathrm{CV} = \frac{s}{|\bar{x}|}$$

WORKED EXAMPLE · APARTMENTS

variable	mean	st. dev.	CV
price	€ 721 993.33	€ 446 306.83	0.6182
number of bedrooms	2.2889	0.6949	0.3036

Price has the far larger SD, but is that just because prices are big numbers? The CV settles it: **price (0.618) is relatively more spread out** than bedrooms (0.304) — about twice as variable in relative terms.

2.4 The other measures of centre & spread

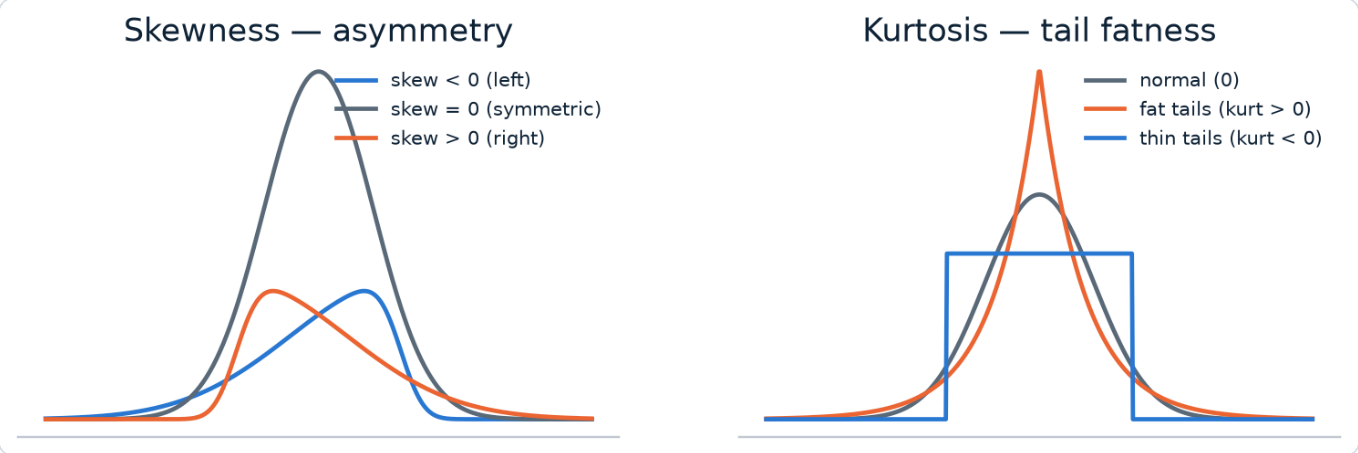
Centre

- **Mean** — the balance point (§2.1).
- **Median** — the middle value: ≈50% of observations below, ≈50% above.
- **Mode** — the most typical value. For binned data it is the centre of the **modal class** = the bin with the **largest density height** (not necessarily the largest count!).

Spread

- **Standard deviation** s (§2.2).
- **Range** = max − min. The full width — very sensitive to extremes.
- **IQR** = Q3 − Q1: the width of the **middle 50%** (see §3).

Shape numbers. Skewness measures asymmetry (0 = perfectly symmetric, positive = tail to the right). **Kurtosis** measures how fat the tails are relative to the normal distribution (0 = like normal, positive = fatter tails).



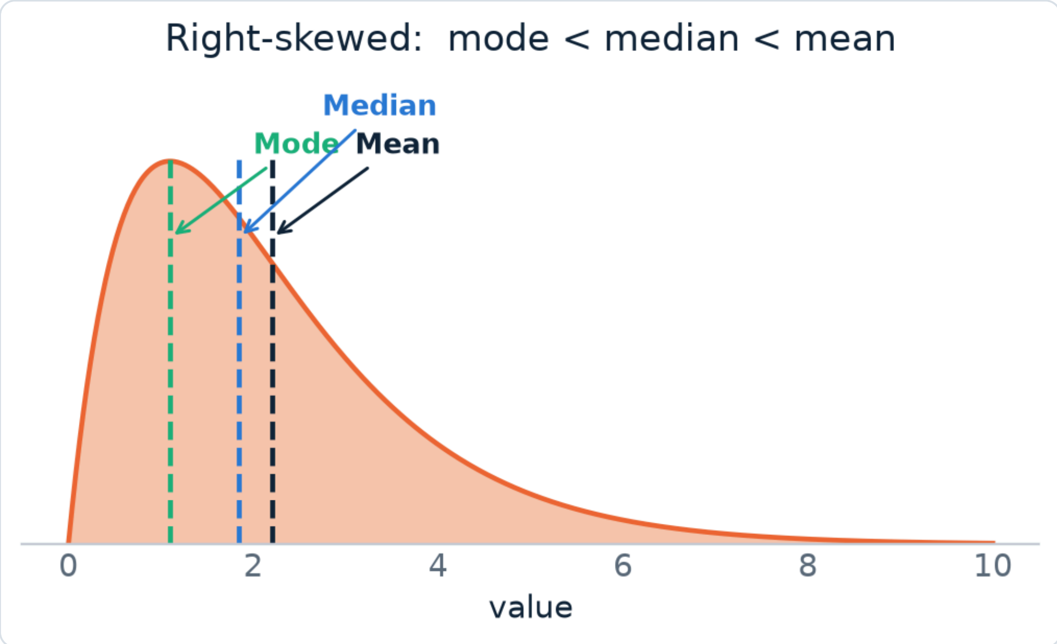
Left: skewness is about the tail direction. Right: kurtosis is about tail thickness.

HOW TO THINK IN THE EXAM — WHICH CENTRE/SPREAD TO USE?

This is the single most-asked conceptual question. Decide with **robustness** (resistance to outliers):

Situation	Centre	Spread	Why
Symmetric, no outliers	Mean	Std. dev.	uses all data efficiently
Skewed / has outliers	Median	IQR	robust — outliers barely move them

Key facts to quote: **mean & SD are heavily pulled by outliers; median, mode & IQR are not.** For a **right-skewed** distribution the **mean** > **median** (the tail drags the mean up), so the **median is usually preferred**. For a symmetric distribution mean ≈ median.



Right-skew ordering: **mode < median < mean**. The long right tail pulls the (non-robust) mean furthest.

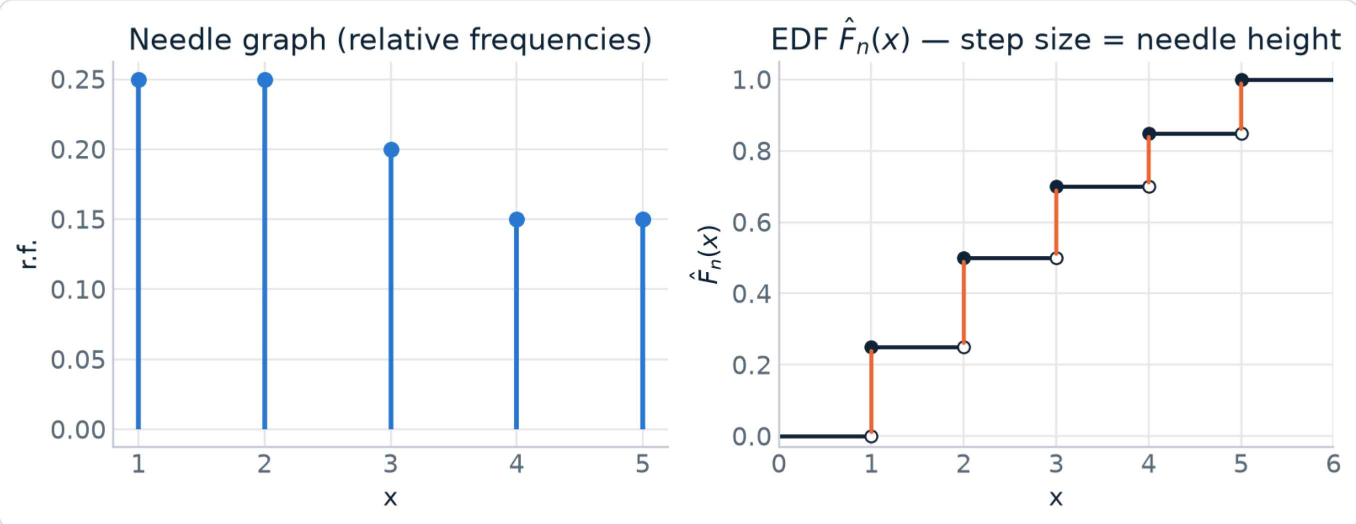
Cumulative distribution, percentiles & the boxplot

3.1 The empirical distribution function (EDF)

The **EDF**, written $\hat{F}_n(x)$, answers a cumulative question for **every** real number x: "what proportion of observations are ≤ x?"

$$\hat{F}_n(x) = \frac{\text{number of observations} \le x}{n}$$

It is just the distribution seen "from below," and it needs **no binning**. Graphically it is a **step function** that starts at 0, jumps up at every observed value, and reaches 1 at the maximum. The **size of each jump equals the relative frequency** (needle height) of that value — that is the exact link between the needle graph and the EDF.



The needle heights (left) become the step sizes of the EDF (right). Closed dot = value included; open dot = not yet.

> **WORKED EXAMPLE · EMAIL DATA (N = 50)**
>
> The cumulative percentage at the outcome 26 255 is 0.90. Read it in words: "**26 255 is a large value in this set — 90% of all outcomes are ≤ 26 255.**" That sentence *is* the meaning of the EDF at a point, and "write the interpretation" questions want exactly this phrasing.

3.2 Percentiles — reading the EDF backwards

A **percentile** runs the EDF in reverse: instead of "value → proportion," you ask "proportion → value." The ***k*-th percentile** is the value below which k% of the ordered data fall. The inverse of the EDF is called the **quantile function** $\hat{Q}(p)$; $\hat{Q}(p)$ is the ***p*-quantile** (so the 0.9-quantile = 90th percentile).

Because the EDF is a step function, "the value at proportion k" is ambiguous, so a convention is needed. **Only the SPSS rule is exam material:**

THE SPSS PERCENTILE RULE — MEMORISE THIS

To find the ***k*-th percentile** of ordered data (n values):

1. Compute the position $(n+1)\cdot k/100$. Let ***i*** = integer part, ***j*** = fractional part.
2. The percentile lies between the i-th and $(i+1)$-th ordered observations:

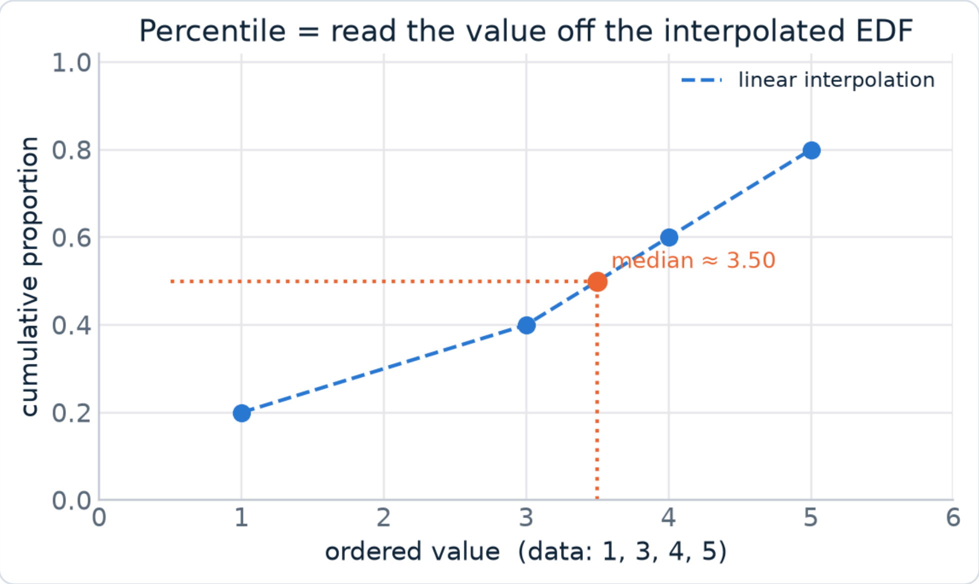
$$\text{percentile} \;=\; x_{(i)} \;+\; j \cdot (\, x_{(i+1)} - x_{(i)} \,)$$

This is just **linear interpolation** between two neighbours.

WORKED EXAMPLE · 90TH PERCENTILE OF THE EMAIL DATA (N = 50)

Position = $(50+1)\cdot 90/100 = 0.9\cdot 51 =$ **45.9** $\rightarrow$ $i = 45$, $j = 0.9$.
So the 90th percentile lies between the 45th and 46th ordered values (26 255 and 26 520):

$$P_{90} = 26\,255 + 0.9\cdot(26\,520 - 26\,255) = 26\,255 + 0.9\cdot 265 = \mathbf{26\,493.5}$$



To get a percentile, go **up** to the interpolated EDF at the target proportion and **across** to the value.

3.3 Median & quartiles are just special percentiles

Quartiles cut the ordered data into four equal parts:

name	= percentile	meaning
Q1 (first quartile)	25th	25% of data below
Q2 = median	50th	the middle
Q3 (third quartile)	75th	75% of data below

The **median** also has a direct definition (which matches the SPSS 50th-percentile rule): if n is **odd** it is the single middle value; if n is **even** it is the average of the two middle values.

WORKED EXAMPLE · MEDIAN & Q1 BOTH WAYS

n = 25 (odd)

median position (26)·0.5 = 13 → **median = 13th value = 3.4**

Q1 position (26)·0.25 = 6.5 → Q1 = 2.1 + 0.5·(0.2) = **2.2**

n = 24 (even)

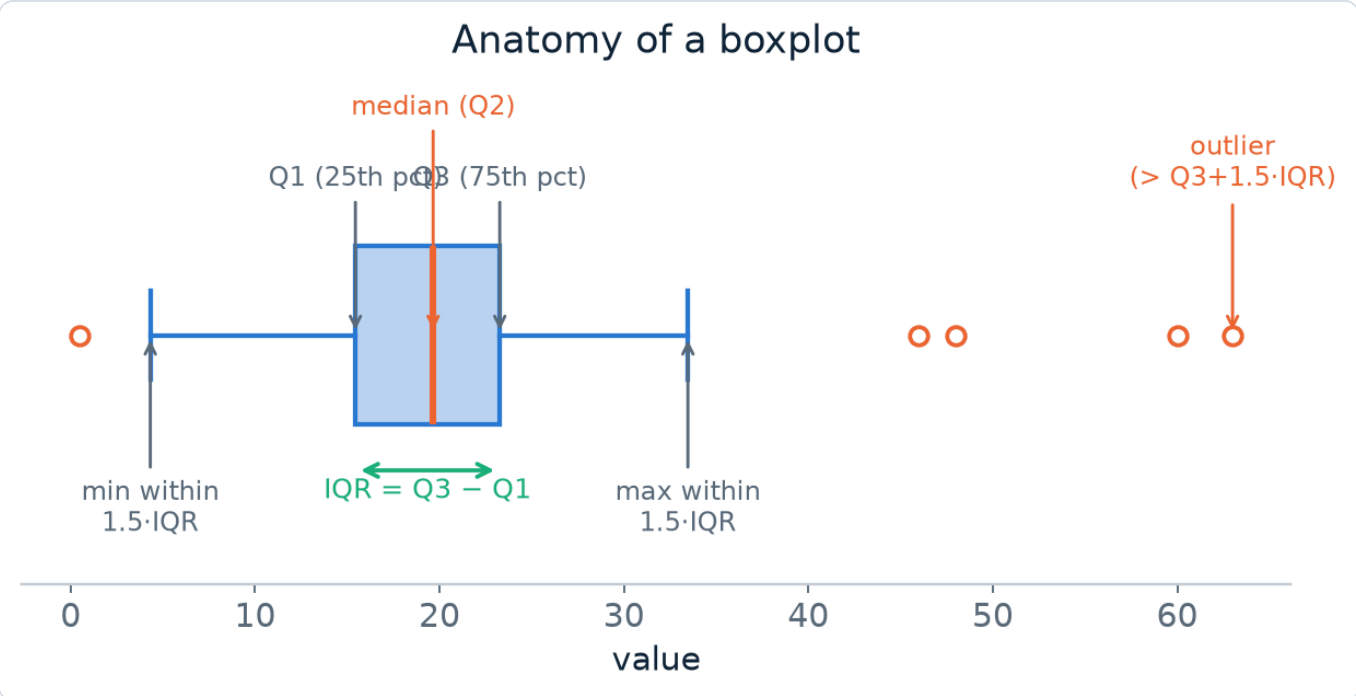
median position (25)·0.5 = 12.5 → median = (3.3+3.4)/2 = **3.35**

Q1 position (25)·0.25 = 6.25 → Q1 = 2.1 + 0.25·(0.2) = **2.15**

3.4 The boxplot — five numbers in one picture

A **boxplot** draws the distribution from just **five summary statistics**: minimum, Q1, median, Q3, maximum.

- The **box** spans Q1 to Q3 — the middle 50% of the data; its width is the **IQR**.
- A **line inside the box** marks the median.
- The **whiskers** reach to the most extreme observations that are still within **1.5·IQR** of the box.
- Anything beyond the whiskers (further than 1.5·IQR below Q1 or above Q3) is drawn as a **separate dot**: a potential outlier.



Every element of a boxplot and the 1.5·IQR fence that defines a potential outlier.

OUTLIERS — THE RULE & THE WARNING

An **outlier** is an observation that looks extreme relative to the rest; the boxplot flags candidates with the **1.5·IQR fence**. Two things the exam wants you to say: **(1)** a flagged point is only a *potential* outlier — investigate it; **(2) never delete an outlier just because it is one** — remove it only with a genuine reason (e.g. a data-entry error). Because outliers can dominate a mean/SD, prefer **robust** statistics (median, IQR) when they are present.

HOW TO THINK IN THE EXAM

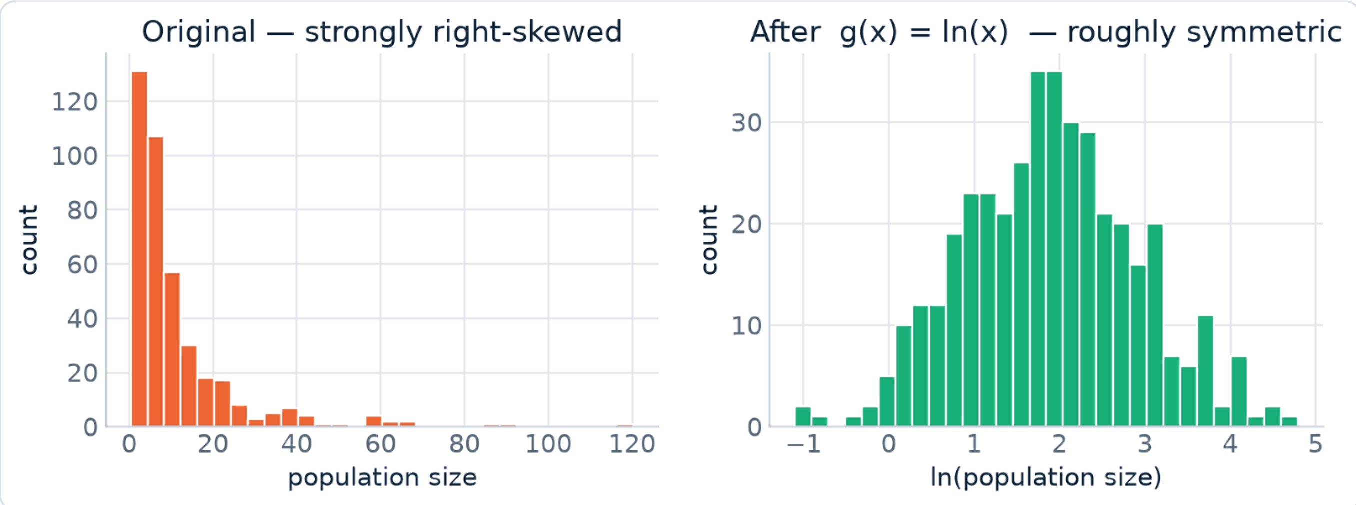
- **"Compute the k-th percentile"** → order the data, apply $(n+1)\cdot k/100$, split into i and j, interpolate. Show the position number.
- **"Interpret $\hat{F}_n$(value)"** → "p% of observations are ≤ value."
- **"Draw / read a boxplot"** → you need the 5 numbers; the box edges are Q1/Q3, IQR = Q3−Q1, and check the 1.5·IQR fence for dots.
- Median vs mean question → tie back to skew (§2.4).

Transformations & the z-score

A **transformation** replaces each observation x_i by $x^*_i = g(x_i)$. We do this either to **fix a skewed shape** or to **change units / standardise**.

4.1 Logarithmic transformation — taming right skew

When a variable is strongly **skewed to the right** because of a few huge values (population size, income), the log $g(x) = \ln(x)$ compresses the large values much more than the small ones, pulling the distribution toward **symmetry**.



Country population size: two extreme values (India, China) create heavy right skew; ln(x) makes the shape roughly symmetric.

4.2 Linear transformation — changing units

A **linear** transformation $g(x) = a + b \cdot x$ (e.g. °C → °F, € → $) shifts and rescales. Its effect on the summary measures is completely predictable — **this is a favourite exam calculation**:

transform	measures of centre (mean, median)	measures of spread (SD, IQR)	variance
shift: $x^* = x + a$	increase by **a**	**unchanged**	unchanged
scale: $x^* = b \cdot x$	multiplied by **b**	multiplied by **|b|**	multiplied by **b^2**

THE INTUITION (SO YOU NEVER MEMORISE IT WRONG)

Shifting slides the whole dataset along the axis — the middle moves, but distances between points don't change, so **spread is untouched**. **Scaling** stretches everything from 0 — both the centre and the distances scale. Spread uses **|b|** (spread can't be negative); variance is squared so it uses **b^2**.

4.3 The z-score — a special, universal linear transform

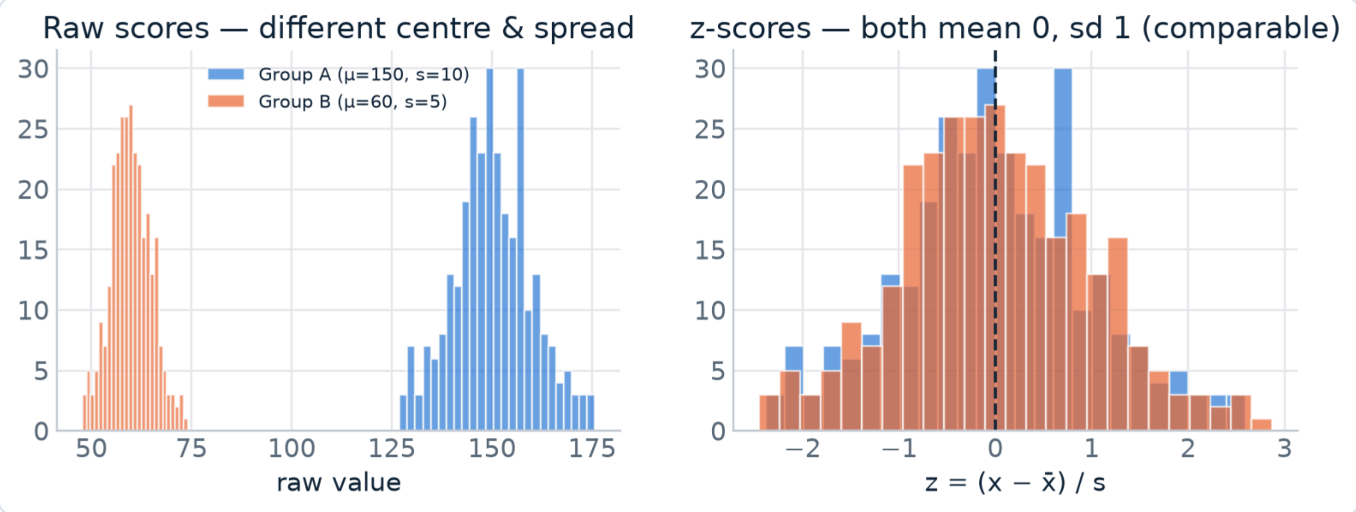
The **z-score** subtracts the mean and divides by the SD:

$$z = \frac{x - \bar{x}}{s}$$

It answers: “**how many standard deviations is this point above (+) or below (−) the mean?**” Applying it to every observation **standardises** the variable. Because it is the linear transform with a = $-\bar{x}/s$ and b = $1/s$, the rules in §4.2 force:

$$\bar{z} = 0 \quad \text{and} \quad s_z = 1$$

Every standardised variable has mean 0 and SD 1, which is exactly why z-scores make **different variables or groups comparable** — a z of +2 means “2 SDs above average” whether the raw data were exam marks or euros.



Two groups on different scales (left) line up on one common scale after standardising (right).

HOW TO THINK IN THE EXAM

- “New mean / SD after $a + bx$?” → apply the table. Watch: spread ignores a, uses $|b|$ (not b); variance uses b^2.
- “Which of two results is relatively better?” across different tests → compute **z-scores** and compare.
- “Why standardise?” → to compare across variables/groups; standardised data always have $\bar{z}$=0, s_z=1.

SECTION 5 · SLIDE SET D2.5

Working from binned data only

Sometimes you never see the raw numbers — only a **frequency table of bins** (e.g. income reported in brackets). You can **still** produce every descriptive statistic, using **one key assumption**:

THE BINNED-DATA ASSUMPTION

Observations are spread out uniformly within each bin. This single assumption justifies both the **linear interpolation** for cumulative frequencies/percentiles and the **bin-midpoint approximations** for the mean and variance below.

5.1 Cumulative frequencies by interpolation

Build a table with absolute (n_i), relative (f_i) and cumulative (F_i) frequencies at the **bin edges**. Between edges, the cumulative curve is drawn as a **straight line** (that is the uniform-within-bin assumption).

WORKED EXAMPLE · HOUSEHOLD INCOME, F(1200)

At the edges: F(1000) = 0.25 and F(1500) = 0.50. The point 1200 is (1200−1000)/(1500−1000) = **0.4** of the way through the bin, so:

$$F(1200) = 0.25 + 0.4 \cdot (0.50 - 0.25) = 0.25 + 0.10 = \mathbf{0.35}$$

“About **35% of household incomes are ≤ €1200**.”

5.2 Percentiles by interpolation (EDF backwards)

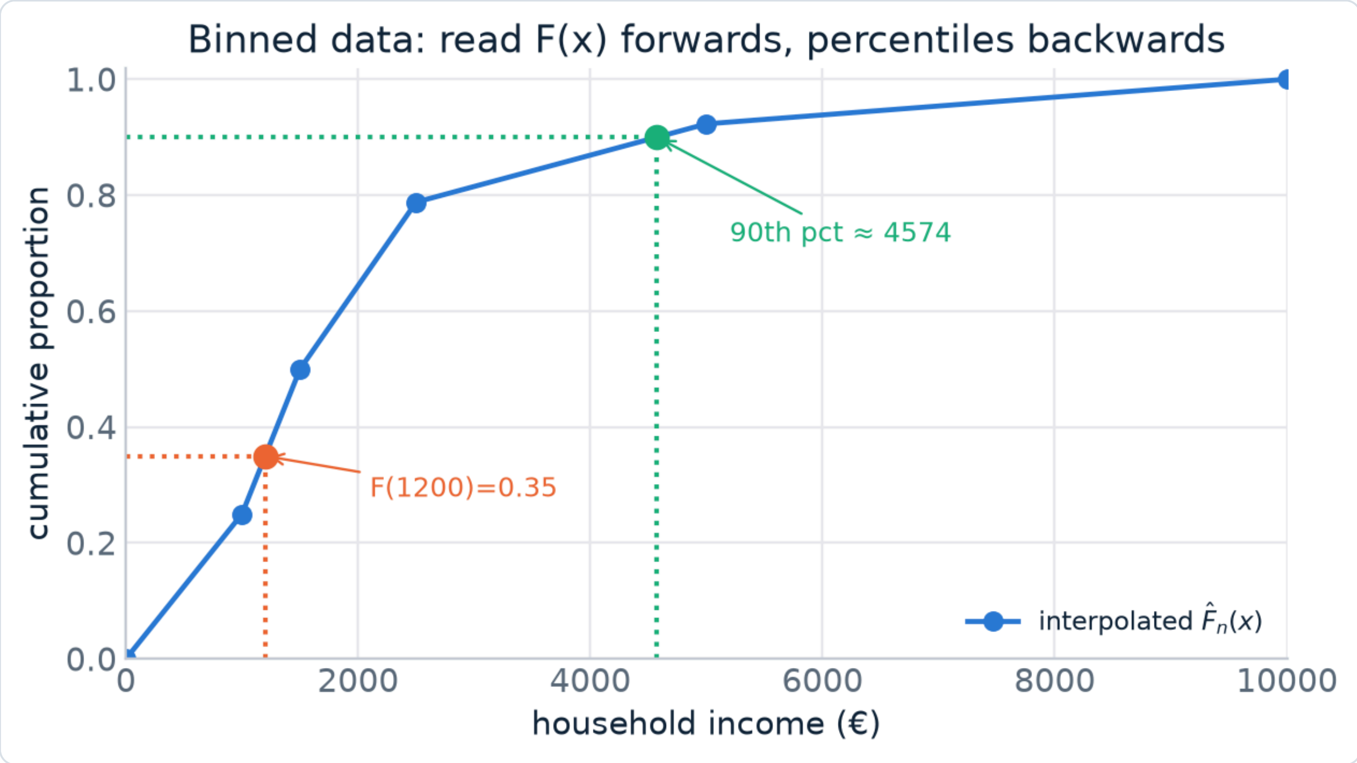
To get a percentile you invert the same interpolated curve: find which bin contains the target proportion, then interpolate for the value.

WORKED EXAMPLE · HOUSEHOLD INCOME, 90TH PERCENTILE

We need the value where F = 0.90. From the table F(2500) = 0.788 and F(5000) = 0.923, so the 90th percentile is in the (2500, 5000] bin. It is (0.90−0.788)/(0.923−0.788) = 0.112/0.135 = **0.82963** of the way in:

$$P_{90} = 2500 + 0.82963 \cdot (5000 - 2500) = 2500 + 2074.07 = \mathbf{4574.07}$$

“About **90% of household incomes are ≤ €4574**.”



Same interpolated curve, two directions: value → proportion gives F(1200)=0.35; proportion → value gives P_{90}≈4574.

5.3 Mean & variance from bins (midpoint approximation)

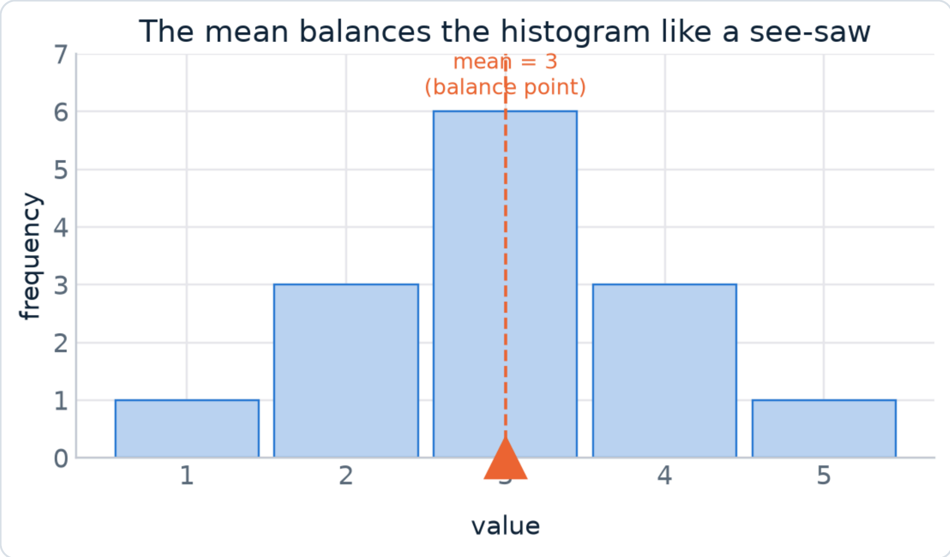
Treat every observation in a bin as if it sat at the **bin midpoint** m_i, then use the discrete formulas from §2:

$$\bar{x} \approx \frac{1}{n} \sum n_i\, m_i \quad \cdot \quad s^2 \approx \frac{1}{n-1} \sum n_i (m_i - \bar{x})^2$$

These are **approximations** ("≈") precisely because the exact positions inside each bin are unknown.

NICE INTUITION · THE MEAN IS THE BALANCE POINT

Picture the density histogram as a physical shape resting on a see-saw. The **mean is the point where it balances**. For a **symmetric** distribution the balance point is dead centre — which is why, for symmetric data, mean = median = centre without any calculation.



The mean is the fulcrum that balances the histogram.

PART II

The relationship between two variables

Part I described one variable at a time. Now we ask whether two variables are **related** — do they move together? The right tool again depends on the **types** of the two variables:

Variable A	Variable B	Graph	Summary	Section
qualitative	qualitative	segmented (100%) bar	contingency table of proportions	6
qualitative	quantitative	side-by-side boxplots	mean / median per group	7
quantitative	quantitative	scatterplot	covariance & correlation	8

Two qualitative variables

To study two categorical variables we cross-tabulate them in a **contingency table** and picture the result with a **segmented bar plot**. Course example: does a person's **handiness** affect whether home maintenance is done **Formally**, **Informally** (off the books) or **DIY**?

6.1 The contingency table

Each cell is the **count** of one combination of outcomes. The **row and column totals ("marginals")** describe each variable **on its own** — they tell you nothing about the *relationship*.

	Clumsy	Neutral	Handy	Very handy	Total
Formal	9	59	76	31	175
Informal	9	37	33	6	85
DIY	8	44	116	83	251
Total	26	140	225	120	511

6.2 Three kinds of proportions — don't mix them up

Raw counts are hard to compare across groups of different sizes, so we convert to proportions. There are **three** ways to divide, and they answer different questions:

divide each count by…	called	each row/col sums to	answers
grand total n	overall proportions	whole table = 100%	"what fraction of everyone is (DIY & handy)?"
its row total	row proportions	each **row** = 100%	"among the Formal group, how is handiness split?"
its column total	column proportions	each **column** = 100%	"among Handy people, how is the mode of work split?"

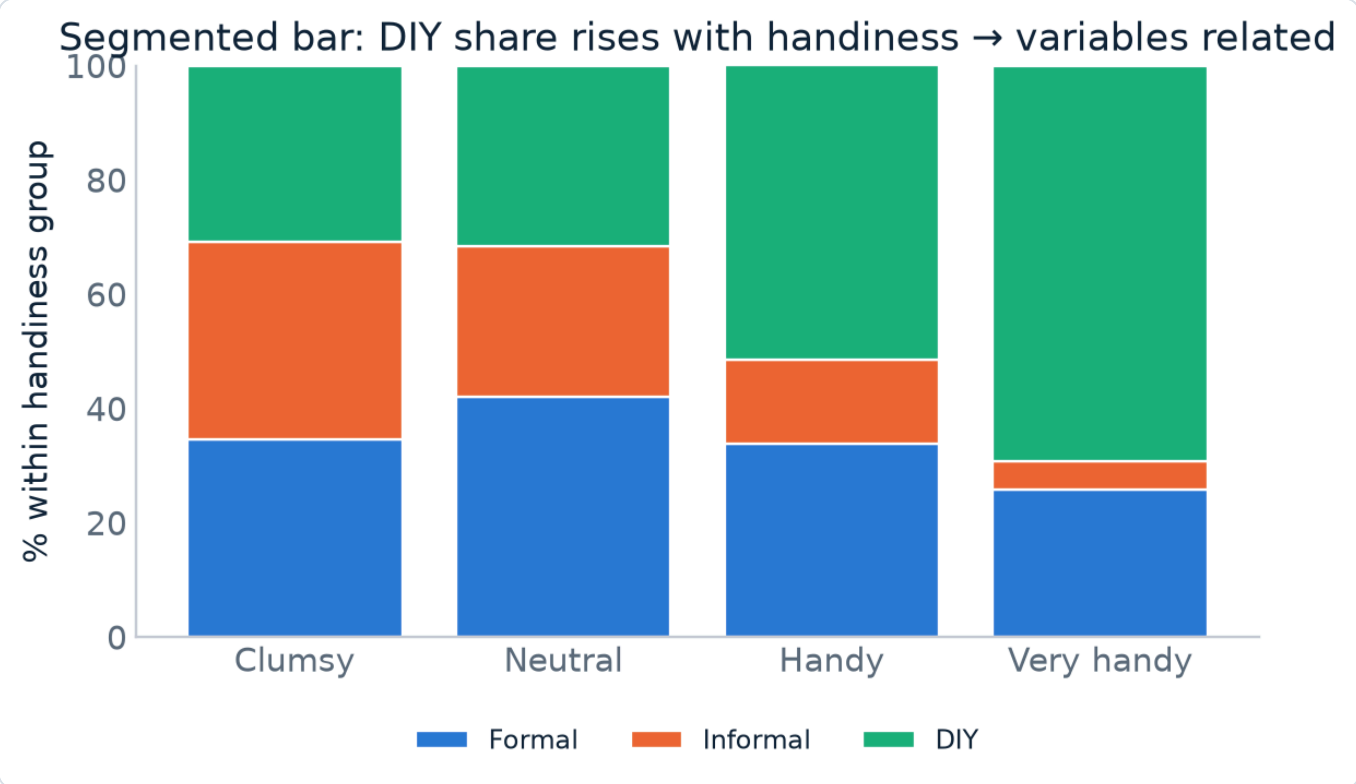
Example column-proportion cell: of the 225 *Handy* people, 116 do DIY → 116/225 = **51.6%**.

6.3 The segmented (100%) bar plot

Put one variable on the x-axis. For each group, stack the **proportions** of the other variable to fill a 100%-tall bar. The reading rule is simple and is the whole point of the graph:

THE DEPENDENCE RULE

If the coloured proportions **stay the same** across the groups on the x-axis, the variables are **independent** (unrelated). If the proportions **change** from group to group, the variables are **dependent** (related).



The **DIY** share climbs from ~31% (Clumsy) to ~69% (Very handy) while **Informal** shrinks — the proportions change, so **handiness and mode of work are related**.

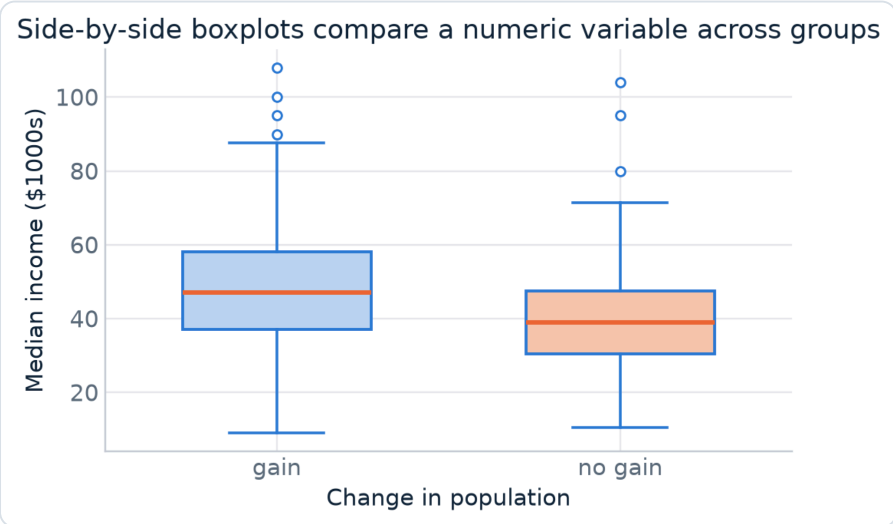
HOW TO THINK IN THE EXAM

- Told to build a proportion table → **check whether they want row, column or overall** proportions; it changes every number. “Given handiness, …” ⇒ column proportions.
- “Are the variables related?” from a segmented bar → look whether the coloured bands keep the same heights across bars. Changing = dependent.
- Remember marginals describe single variables only.

One qualitative + one quantitative variable

Here we compare a **numeric** variable **across the groups** of a categorical variable — e.g. is the **poverty / median-income** level different for counties that **gained** vs **did not gain** population?

- **Graphically:** draw a **boxplot for each group on the same scale** — **side-by-side boxplots**. Because they share an axis you can directly compare centre, spread and skew.
- **Numerically:** report the **mean (or median), SD, … separately for each group**, then compare.



Side-by-side boxplots: the “gain” group sits higher (larger median income) and both groups have high outliers.

HOW TO THINK IN THE EXAM

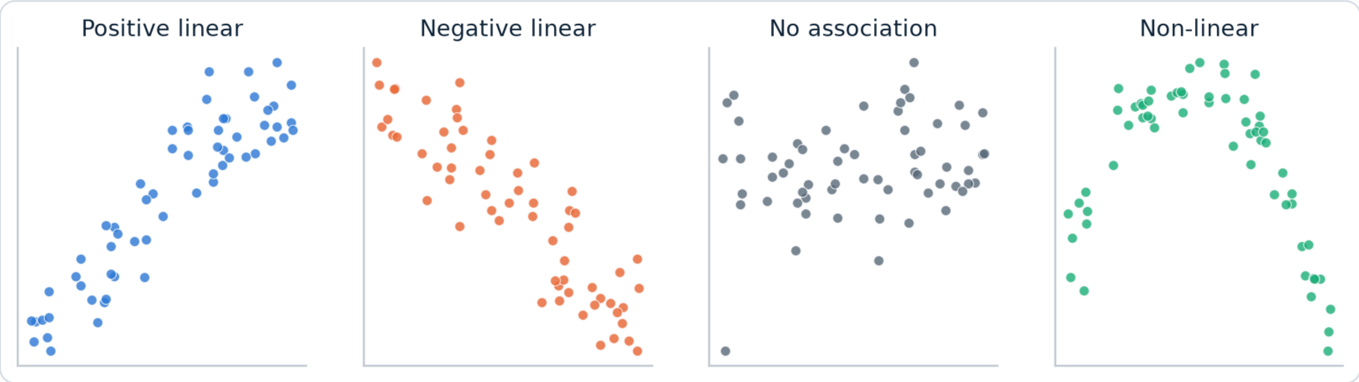
Compare the groups on **three** things: **centre** (which median/mean is higher?), **spread** (which box/IQR is wider?), and **shape/outliers** (skew, separate dots). A relationship exists if the numeric variable’s **distribution shifts between groups**. Prefer median/IQR language when boxes show outliers.

Two quantitative variables

8.1 The scatterplot

Plot the pair (x_i, y_i) for every respondent. The cloud of points reveals three things at a glance:

- **Direction:** positive (both rise together) or negative association.
- **Strength:** tight cloud (strong) vs diffuse cloud (weak).
- **Form:** linear (a straight-line pattern) vs non-linear (curved). Some curves can be straightened with a transform.



Direction, strength and form. "Linear" means the points scatter *around* a line — they need not lie exactly on it.

ASSOCIATION ≠ CAUSATION

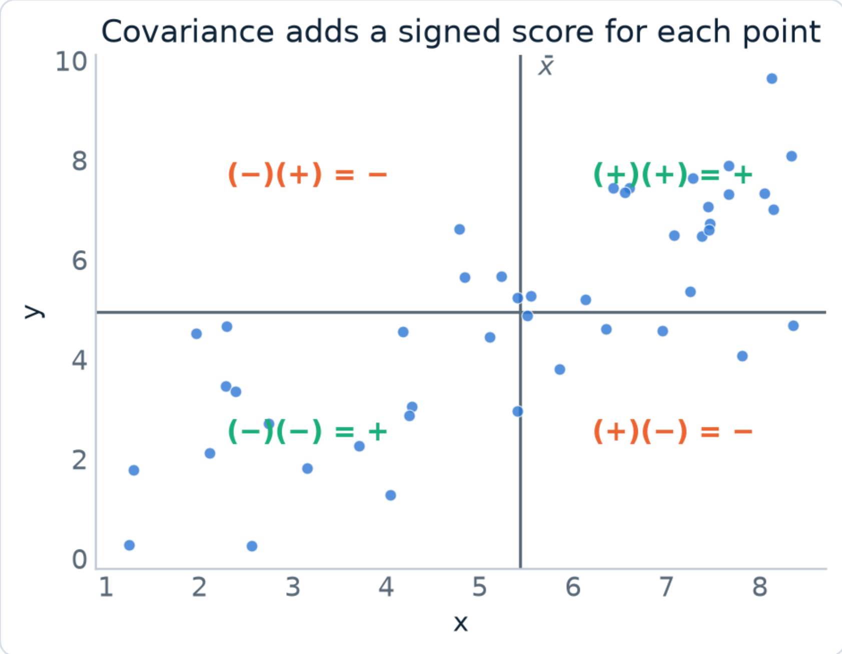
A scatterplot can only reveal that two variables **move together**; it cannot prove one **causes** the other (that needs an experiment, not an observational study). Keep this sentence ready — it is a standard exam point.

8.2 Covariance — a signed measure of joint movement

For each point, compute the product of its deviations from the two means, $(x_i-\bar{x})(y_i-\bar{y})$. Points where both variables are on the same side of their mean give a **positive** contribution; opposite sides give a **negative** one. Averaging these gives the **covariance**:

$$\boxed{\mathrm{Cov}(x,y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}$$

Note $\mathrm{Cov}(x,x) = s_x^2$ — covariance of a variable with itself is just its variance.



The mean lines split the plane into four quadrants; the sign of each point's contribution depends on its quadrant.

WHY COVARIANCE ALONE ISN'T ENOUGH

Covariance gets the **direction** right (+ / −) but its **size is not interpretable**: it changes if you change the units (measure income in cents vs euros and the covariance explodes). We need a **unit-free** version.

8.3 Correlation — standardised covariance

Standardise each variable first (§4.3), then take the covariance. Equivalently, divide covariance by the two standard deviations. The result is the **correlation coefficient** r:

$$r = \frac{1}{n-1} \sum \frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y} = \frac{\mathrm{Cov}(x,y)}{s_x\, s_y}$$

Key properties of *r*

- Always **−1 ≤ *r* ≤ +1**.
- Sign = direction; magnitude = **strength of the linear association.**
- **Unit-free**: unchanged by any linear transformation of x or y.
- **Sensitive to outliers.**

Interpreting |*r*|

|r|	linear strength
< 0.3	very weak
0.3 – 0.5	weak
0.5 – 0.7	moderate
0.7 – 0.85	strong
0.85 – 0.95	very strong
> 0.95	extremely strong

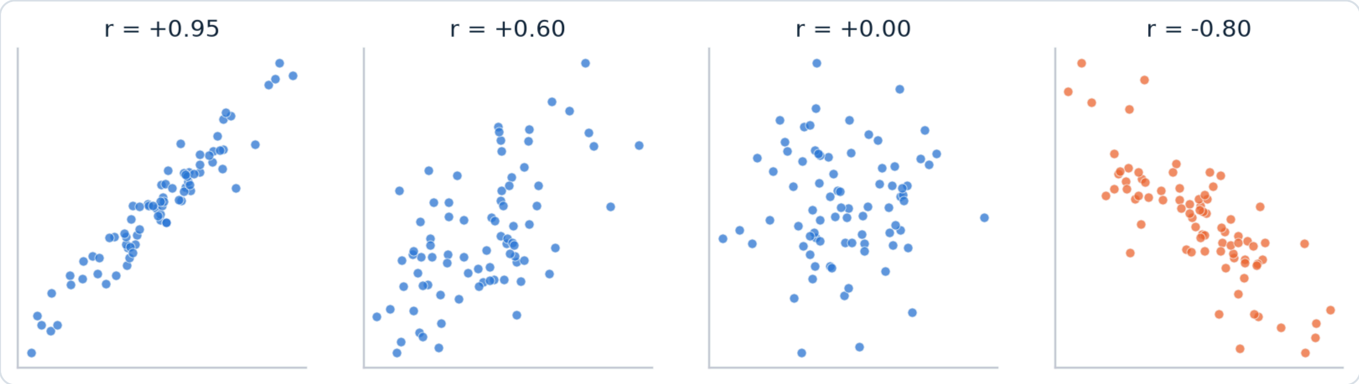
WORKED EXAMPLE · APARTMENTS (AREA VS PRICE)

n = 45. Price: $\bar{y}$ = 721 999.33, s_y = 446 306.83. Area: $\bar{x}$ = 111.16, s_x = 36.969. The result is **r(x,y) = 0.716** → a **strong, positive, linear** association: bigger apartments tend to cost more.

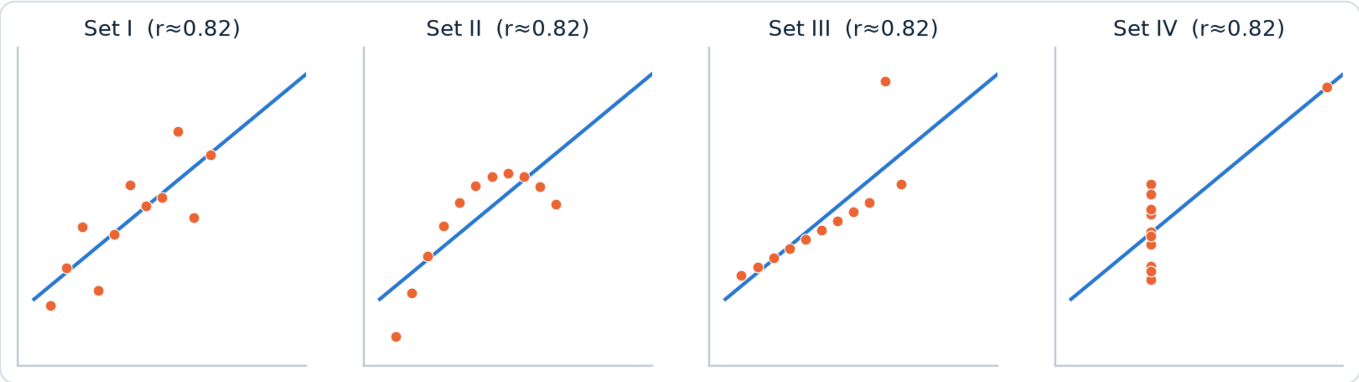
THE TWO CORRELATION TRAPS (BOTH ARE EXAM FAVOURITES)

1 · *r* only measures linear strength. A perfect U-shape can have $r \approx 0$ even though the variables are perfectly related. **Always plot the data with the correlation.**

2 · Anscombe's quartet: four completely different datasets can share the **same $r = 0.816$** — one linear, one curved, one a line dragged by a single outlier, one vertical with one leverage point. A number never replaces the graph.



What different *r* values look like — sign is direction, magnitude is tightness.



Anscombe's quartet: identical $r \approx 0.82$ (and identical means, SDs, regression line), utterly different realities.

8.4 When a scatterplot isn't the best picture

- **Time plot** — if x is time, connect points in time order to show the trend (e.g. monthly unemployment rate).
- **Mapping plot** — if there is a spatial component, shade a map (e.g. poverty rate by county).
- **Bubble graph** — a scatterplot can't show that the **same pair occurs several times**; a bubble graph encodes frequency as the circle's **area** (double the area = observed twice as often).

Linear combinations of variables

A **linear combination** builds a new variable from others, e.g. a portfolio's gain $G = a + b \cdot X + c \cdot Y$. The exam wants its **mean** and **spread**. There are two cases.

9.1 One variable: *a + bX* (recap of §4.2)

$$\overline{a+bx} = a + b \cdot \bar{x} \quad \cdot \quad s_{a+bx} = |b| \cdot s_x \quad \cdot \quad \mathrm{var}(a+bx) = b^2 \cdot \mathrm{var}(x)$$

WORKED EXAMPLE · DAVID'S SINGLE STOCK

Gain $G = 4000X - 50$, with $\bar{x} = 0.021$, $s_x = 0.08486$.

mean gain = $-50 + 4000 \cdot (0.021)$ = **\$34**

$s_G = 4000 \cdot 0.08486$ = **\$339.44** (the −50 shift does not affect spread)

9.2 Two variables: *a + bX + cY* — the important formula

The mean is easy (means just add through). The spread is where students slip, because of the **cross term**:

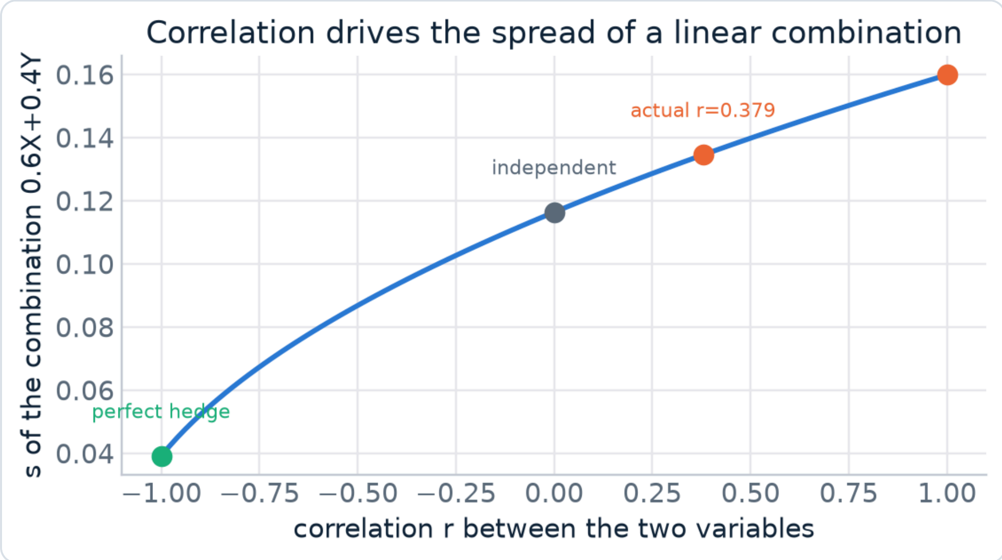
$$\overline{a+bx+cy} = a + b \cdot \bar{x} + c \cdot \bar{y}$$

$$\boxed{\mathrm{var}(a+bx+cy) = b^2\mathrm{var}(x) + c^2\mathrm{var}(y) + 2bc \cdot s_x s_y \cdot r}$$

READ THE CROSS TERM — IT IS THE WHOLE POINT

The $\mathbf{2bc \cdot s_x s_y \cdot r}$ term (= $2bc \cdot \mathrm{Cov}(x,y)$) is the **only** place the relationship between X and Y enters.

- **Independent (r = 0):** the cross term vanishes → variances simply add: var = $b^2\mathrm{var}(x) + c^2\mathrm{var}(y)$.
- **Positively correlated (r > 0):** risk is *larger* than if independent.
- **Negatively correlated (r < 0):** the variables partly cancel → risk is *smaller* (the idea behind diversification / hedging).



Same weights, same individual SDs — yet the spread of 0.6X+0.4Y depends entirely on r.

WORKED EXAMPLE · LEONARD, TWO INDEPENDENT STOCKS

$G = 6000X + 2000Y$; $\bar{x} = 0.021$, $s_x = 0.08486$; $\bar{y} = 0.004$, $s_y = 0.051962$; **independent → r = 0**.

mean = 6000·0.021 + 2000·0.004 = **$134**

$s_G^2 = 6000^2 \cdot 0.08486^2 + 2000^2 \cdot 0.051962^2 = 270\,044.1 \rightarrow s_G =$ **$519.66**

WORKED EXAMPLE · SANDRA, TWO CORRELATED STOCKS

Return $R = 0.6X + 0.4Y$; $\bar{x} = 0.0202$, $s_x = 0.1659$; $\bar{y} = 0.0175$, $s_y = 0.1509$; **r = 0.379**.

mean = 0.6·0.0202 + 0.4·0.0175 = **0.01912**

$s_R^2 = 0.6^2 \cdot 0.1659^2 + 0.4^2 \cdot 0.1509^2 + 2\cdot(0.6)(0.4)(0.1659)(0.1509)(0.379) = 0.018106$

$s_R =$ **0.1346** — note you must include the +cross term because r ≠ 0.

HOW TO THINK IN THE EXAM

1. Write the combination in $a + bX + cY$ form; read off a, b, c.
2. Mean: plug means in directly.
3. Variance: **always start from the full formula with the cross term.** Only drop it if you are told the variables are **independent / uncorrelated** (r = 0).
4. SD = √variance. Keep units ($, %, decimal) consistent.

REFERENCE

Formula cheat-sheet & exam decision guide

Every formula in one place

quantity	formula	notes
Mean	$\bar{x} = (1/n)\sum x_i = \sum \hat{p}_i m_i$	centre; not robust
Variance	$s^2 = (1/(n-1))\sum(x_i-\bar{x})^2$	squared units; ÷(n−1) = df
Std. deviation	$s = \sqrt{s^2}$	original units; not robust
Variation coeff.	$CV = s / \lvert\bar{x}\rvert$	unit-free; compare spreads
Density height	$h_i = \hat{p}_i / b_i$	area = proportion; ∑area = 1
EDF	$\hat{F}_n(x) = (\#obs \le x)/n$	step function; jump = r.f.
Percentile (SPSS)	$pos = (n+1)k/100 = i.j \to x_{(i)} + j(x_{(i+1)}-x_{(i)})$	linear interpolation
IQR	IQR = Q3 − Q1	middle 50%; robust
Outlier fence	below Q1−1.5·IQR or above Q3+1.5·IQR	boxplot dots
z-score	$z = (x-\bar{x})/s$	$\Rightarrow \bar{z}=0,\ s_z=1$
Linear transform	mean: $a+b\bar{x}$; SD: $\lvert b\rvert s$; var: $b^2\cdot var$	shift doesn't change spread
Covariance	$Cov = (1/(n-1))\sum(x_i-\bar{x})(y_i-\bar{y})$	signed; units matter
Correlation	$r = Cov(x,y) / (s_x s_y)$	−1…1; linear only; not robust
Lin. combination	mean = $a+b\bar{x}+c\bar{y}$; var = $b^2 var(x)+c^2 var(y)+2bc\cdot s_x s_y\cdot r$	drop cross term only if r=0

Decision guide — "what should I use?"

The question asks…	Reach for…
Show the shape of one discrete variable	needle graph → describe symmetry/modality/outliers
Show the shape of one continuous variable	histogram (density if bins unequal)
Typical value & spread, no outliers/symmetric	mean & standard deviation
Typical value & spread, skewed/outliers	median & IQR (robust); boxplot
Position of a value / a percentile	EDF & SPSS interpolation
Compare spreads across different units	variation coefficient, or z-scores
Make a right-skewed variable symmetric	log transformation
Relate two categorical variables	contingency table (row/col proportions) + segmented bar
Relate a category to a number	side-by-side boxplots + group means
Relate two numbers	scatterplot + correlation r (always plot too!)
Mean/risk of a weighted sum of variables	linear-combination formulas (mind the cross term)

TEN SENTENCES THAT ANSWER MOST CONCEPTUAL QUESTIONS

1. Skew points toward the **long tail**; right-skew ⇒ mean > median.
2. On a density histogram, **proportion = area**, not height; total area = 1.
3. Divide by **n−1** (degrees of freedom) so s^2 is unbiased.
4. Interpret spread with **SD** (same units), never the variance.
5. Mean & SD are **not robust**; median & IQR are.
6. A boxplot dot is a **potential** outlier — investigate, don't delete.
7. z-scores make different variables **comparable** (mean 0, SD 1).
8. Segmented bars: bands **change ⇒ dependent**; stay same ⇒ independent.
9. **Correlation ≠ causation**, and r only measures **linear** strength — always plot.
10. In a linear combination, the **cross term** (via r) is the only place the relationship enters.

Study guide compiled from course slide sets D2.1–D2.5 and D3.1–D3.4 (Descriptive Statistics — Quantitative variables & Relationships between variables). Figures are clean re-creations of the slides' graphs, drawn to the same examples and values so you can map each one back to your deck.